

DATA STRUCTURE LAB

# Array ArrayList

// [] [] [] [] [] [] [] [] [] [] [] [] [] [] [] []

**length**

□ □ □ □ □

**size ≠ capacity**

□ □ □ □ □ □ □ □ ≠ □ □ □ □ □



# API

```
1 int[] scores = {85, 92, 78};  
2 scores[1] = 95;  
3 int n = scores.length;
```

NO GRAPH  
NO GRAPH  
NO GRAPH  
NO GRAPH  
[ ] · NO GRAPH  
NO GRAPH  
NO GRAPH  
NO GRAPH

```
1 List<Integer> scores = new ArrayList<>();  
2 scores.add(85);  
3 scores.set(0, 95);  
4 int n = scores.size();
```

NO GRAPH  
NO GRAPH  
API · □□□□□

size NO GUPH capacity ☐☐☐☐☐☐

10

20

30

☐

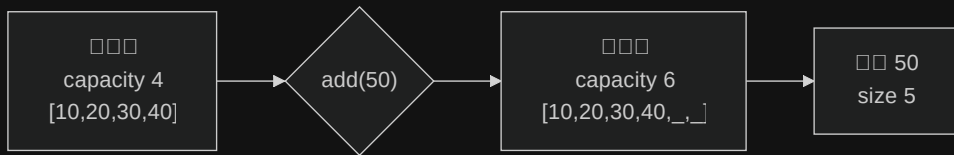
size = 3

☐☐☐☐☐☐☐☐

capacity = 4

☐☐☐☐☐☐☐☐

capacity NO GUPH NO GUPH NO GUPH NO GUPH NO GUPH NO GUPH NO GUPH NO GUPH List API ☐☐☐☐☐☐



**4 → 6**  
 $4 + \text{floor}(4 / 2)$

**4**  
add  $O(n)$

**0(1)**  
□□□□□□

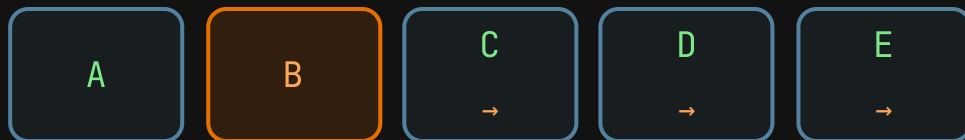
OpenJDK 1.5× List □□□□



---

```
1 List<Order> orders = new ArrayList<>(incoming);
```

```
□□□□□□□□□□□□□□□□□□□□□□
```



**add**

□□□□□□□□

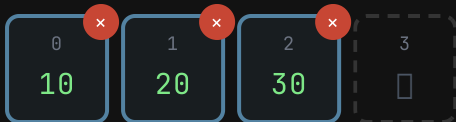
**index 1** □□

□□□□□□□□ $O(n)$



Array length

ArrayList size / capacity



size = 3 · capacity = 4 · 0 · 0

40

add

1

insert

□

□□□□□□□□

□□□ capacity □ add□□□ insert□remove□□□□□□□□□□

# primitive wrapper

```
1 int[] raw = {1, 2, 3};  
2 // 000 int
```

Array NOT ES6 NOT ES6 NOT ES6 NOT ES6 primitive NOT ES6 int NOT ES6 double NOT ES6 char

```
1 List<Integer> boxed = new ArrayList<>();  
2 boxed.add(1); // int → Integer  
3 int n = boxed.get(0); // Integer → int
```

NOT ES6 NOT ES6 NOT ES6 NOT ES6 NOT ES6 NOT ES6 List<int> NOT ES6 NOT ES6 wrapper NOT ES6  
autoboxing





```
1 List<Integer> values = new ArrayList<>();
2 values.add(null);
3
4 Integer boxed = values.get(0); // boxed == null
5 int raw = values.get(0);      // NullPointerException
6
7 Integer maybe = values.get(0);
8 if (maybe != null) {
9     int safe = maybe;
10    System.out.println(safe);
11 }
```

values.get() returns Integer → int

# Arrays.asList

```
1 List<String> readOnly = List.of("Ada", "Linus");  
2 // set / add
```

```
Arrays.asList().view().asArrayList()
```



```
import java.util.*;
```

```
class Roster {  
    public static void main(String[] args) {  
        List<String> names =  
            new ArrayList<>(List.of("Ada", "Linus"));  
        names.add("Grace");  
        System.out.println(names);  
    }  
}
```

□□

1. `String[]`
2. `ArrayList`
3. `Monaco` `Java`



| <div>□□</div>                                                                                                                   | Array                                                                                                    | ArrayList                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <code>get(index)</code>                                                                                                         | $O(1)$                                                                                                   | $O(1)$                                                             |
| <code>set(index)</code>                                                                                                         | $O(1)$                                                                                                   | $O(1)$                                                             |
| <div><div>NO ERROR</div><div>NO ERROR</div><code>add</code></div>                                                               | □□□                                                                                                      | <div><div>NO ERROR</div><div>NO ERROR</div><math>O(1)</math></div> |
| <div><div>NO ERROR</div><div>NO ERROR</div><div>NO ERROR</div><div>NO ERROR</div>//<div>NO ERROR</div><div>NO ERROR</div></div> | <div><div>NO ERROR</div><div>NO ERROR</div><div>NO ERROR</div><div>NO ERROR</div><math>O(n)</math></div> | $O(n)$                                                             |
| □□                                                                                                                              | □□                                                                                                       | <div><div>NO ERROR</div><div>NO ERROR</div><math>O(n)</math></div> |

Big-O □□□□□□boxing□□□□□□□□□□□□



□□□□□□

  7  primitive □□

double[]

□□□□

□□□□□□□

ArrayList

  8 × 8

□□□□□□□□

char[][]

□□□□

□□□□□□

ArrayList

   primitive   Array              ArrayList□



size ☐☐☐☐☐ capacity ☐☐☐☐☐

☐ ☐ ☐ 1.5x ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

primitive ☐ ☐ ☐ ☐ ☐ ☐ ☐ API ☐☐☐

